

For the record, we'll state a formal version of this result. A random variable like C that counts steps to first failure is said to have a *geometric distribution* with parameter p .

Definition 18.4.6. A random variable, C , has a *geometric distribution* with parameter p iff $\text{codomain}(C) = \mathbb{Z}^+$ and

$$\Pr[C = i] = (1 - p)^{i-1} p.$$

Lemma 18.4.7. If a random variable C has a *geometric distribution* with parameter p , then

$$\text{Ex}[C] = \frac{1}{p}. \quad (18.8)$$

18.4.7 Expected Returns in Gambling Games

Some of the most interesting examples of expectation can be explained in terms of gambling games. For straightforward games where you win w dollars with probability p and you lose x dollars with probability $1 - p$, it is easy to compute your *expected return* or *winnings*. It is simply

$$pw - (1 - p)x \text{ dollars.}$$

For example, if you are flipping a fair coin and you win \$1 for heads and you lose \$1 for tails, then your expected winnings are

$$\frac{1}{2} \cdot 1 - \left(1 - \frac{1}{2}\right) \cdot 1 = 0.$$

In such cases, the game is said to be *fair* since your expected return is zero.

Splitting the Pot

We'll now look at a different game which is fair—but only on first analysis.

It's late on a Friday night in your neighborhood hangout when two new biker dudes, Eric and Nick, stroll over and propose a simple wager. Each player will put \$2 on the bar and secretly write "heads" or "tails" on their napkin. Then you will flip a fair coin. The \$6 on the bar will then be "split"—that is, be divided equally—among the players who correctly predicted the outcome of the coin toss. Pot splitting like this is a familiar feature in poker games, betting pools, and lotteries.

This sounds like a fair game, but after your regrettable encounter with strange dice (Section 16.3), you are definitely skeptical about gambling with bikers. So before agreeing to play, you go through the four-step method and write out the